Thoughts And Rambles On Probability And Statistics

Problem Set 2: Random Variables

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These are the end of chapter problems for Chapter 2 of the notes. Equation numbers refer to the full notes.

1. Let X be such that $\mathbb{P}(X = 2) = \mathbb{P}(X = 3) = \frac{1}{10}$ and $\mathbb{P}(X = 5) = \frac{8}{10}$. Plot the cumulative distribution function $F(x)$. Use F_X to find $\mathbb{P}(2 < X \leq 4.8)$ and $\mathbb{P}(2 \leq X \leq 4.8)$.

2. Let $X \sim N(0, 1)$ and let $Y = e^X$.

(a) Find the PDF for Y. Plot it.

(b) **(Computational)** Generate a vector $\mathbf{x} = (x_1, \ldots, x_{10,000})$ consisting of 10,000 random standard Normal variables. Let $\mathbf{y} = (y_1, \ldots, y_{10,000})$, where $y_i = e^{x_i}$. Draw a histogram of $\mathbf{y}$ and compare it to the PDF you found in part (a).

3. Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$, and assume that X and Y are independent. Show that the distribution of X given that $X + Y = n$ is $\text{Binomial}(n, \pi)$, where $\pi = \frac{\lambda}{\lambda+\mu}$. (**hint:** use the law of total probability for discrete random variables)

4. Let $X, Y \sim \text{Uniform}(0, 1)$ be independent. Find the PDF for $X - Y$ and X/Y.

5. Let $X_1, \ldots, X_n \sim \text{Exp}(\beta)$ be i.i.d. Let $Y = \max\{X_1, \ldots, X_n\}$. Find the PDF of Y.

Hint: $Y \leq y$ if and only if $X_i \leq y$ for $i = 1, \ldots, n$.

6. Show that:

(a) If X is a random variable, then so is $|X|$.

(b) The converse does not necessarily hold.

7. Suppose that X is an integer-valued random variable, and let $m \in \mathbb{N}$. Show that:

$$\sum_{n=-\infty}^{\infty} \mathbb{P}(n < X \leq n + m) = m.$$

8. Let X and Y be random variables and suppose that $A \in \mathcal{F}$. Prove that $Z = X\mathbf{1}_A + Y\mathbf{1}_{A^c}$ is a random variable.